Quality of Life and Clinical Outcome Comparison of Semitendinosus and Gracilis Tendon Versus Patellar Tendon Autografts for Anterior Cruciate Ligament Reconstruction

An 11-Year Follow-up of a Randomized Controlled Trial

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Background: There are still controversies about graft selection for primary anterior cruciate ligament reconstruction. Prospective, randomized long-term studies are needed to determine the differences between the graft materials.

Hypothesis: Eleven years after anterior cruciate ligament reconstruction there is no difference in functional outcome and quality of life between patients with patellar tendon or hamstring tendon autografts; however, the patients with patellar tendon autograft would have a higher prevalence of osteoarthritis.

Study Design: Randomized controlled trial; Level of evidence, 2.

Methods: From June 1999 to March 2000, 64 patients were included in this prospective study. A single surgeon performed primary arthroscopically assisted anterior cruciate ligament reconstruction in an alternating sequence. In 32 patients, anterior cruciate ligament reconstruction was performed with hamstring tendon autograft (semitendinosus and gracilis [STG] group) while in the other 32 patients the reconstruction was performed with patellar tendon autograft (PT group).

Results: At the 11-year follow-up, no statistically significant differences were seen with respect to the Lysholm score and Short Form-36, KT-1000 arthrometer laxity testing, anterior knee pain, single-legged hop test, or International Knee Documentation Committee (IKDC) classification results. Positive pivot-shift test (1+) was significantly more frequent in the PT group (P = .036). Twenty-two patients (81%) in the STG group and 18 patients (72%) in the PT group were still at their preinjury level of activity. Graft rupture occurred in 2 patients from the STG group (6%) and in 4 patients from the PT (12%). Grade B and C abnormal radiographic findings were seen in 84% (21 of 25) of patients in the PT group and in 63% (17 of 27) of patients in the STG group (P = .008).

Conclusion: Both hamstring and patellar tendon autografts provided good subjective outcomes and objective stability at 11 years. Positive pivot-shift test (1+) was significantly more frequent in the PT group. No significant differences in the rate of graft failure were identified. Patients with patellar tendon graft had a greater prevalence of osteoarthritis at 11 years after surgery.

Keywords: anterior cruciate ligament (ACL) reconstruction; hamstring tendons (STG); patellar tendon (PT); pain and ACL; ACL and arthritis; gender laxities; long-term clinical study

The goals of anterior cruciate ligament (ACL) reconstructions are to decrease symptoms, improve function, and return patients to their preinjury level of activity. Anterior cruciate ligament reconstruction is a common procedure that usually allows predictable and timely return to function for the patient. However, an increased incidence of osteoarthritic change has been found after surgical reconstruction as well.[28] A variety of graft sources, such as autografts, allografts, and synthetic grafts, have been used for ACL reconstructions. Because of numerous biologic

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advantages, the most frequently used grafts for intra-articular ACL reconstruction are the autologous patellar tendon[2,3] or doubled semitendinosus and gracilis tendon autografts.[19,34] Disadvantages of patellar tendon autograft are risk of patellar fracture, potential increase in patellofemoral pain, and persistent patellar tendon weakness or rupture.[7,21,22] Disadvantages of hamstring tendon autograft include potential hamstring muscle weakness[8] and slower healing of the graft attachment site.[17,33] Graft fixation is crucial in ACL reconstruction, and it is the weakest link in the initial 6- to 12-week period during which healing of the graft to the host bone occurs. Aperture fixation by interference screws, with either patellar tendon or 4-strand hamstring tendon autograft, provides required tensile load of construct for accelerated postoperative rehabilitation.[24,27,30]

Few prospective long-term studies, and none with longer than 10 years' follow-up, have reported about comparing the knee function and radiologic outcomes after ACL reconstruction using the 2 different autografts (patellar tendon vs 4-strand hamstring tendon autografts). The choice of graft material for ACL reconstruction remains controversial. Well-controlled, prospective long-term studies comparing patellar tendon with hamstring grafts are needed. On one hand, we have studies that find that functional outcome and prevalence of osteoarthritis do not depend on graft choice,[4,14] while on the other hand, some studies find that prevalence of osteoarthritis is higher in patients with patellar tendon autograft.[16,26] The present study is a prospectively randomized, 11-year follow-up study, designed to compare the outcome after arthroscopic single-incision ACL reconstruction with patellar tendon or 4-strand hamstring tendon autograft, both fixed with interference screws by the same surgeon, using the same surgical technique and aggressive postoperative rehabilitation. The present study is a follow-up of a prior 5-year follow-up study.[28] The 5-year follow-up study proved that patients with patellar tendon autografts had greater prevalence of osteoarthritis. The hypotheses of the 11-year follow-up study were that progress of osteoarthritis still has no effect on the functional outcome and quality of life of the patients.

MATERIALS AND METHODS

Patient Selection

In this study, the key indication for surgical reconstruction was a clinical diagnosis of ACL rupture in a patient desiring to return to his or her preinjury level of activity. Patients with associated ligament injury, previous meniscectomy, an abnormality seen radiographically, an abnormal contralateral knee joint, or who did not wish to participate in a research program were excluded from the study. Also, patients with ACL reconstruction on the contralateral knee, or who required revision surgery during the follow-up period, were excluded from the analysis. From June 1999 to March 2000, 64 patients underwent ACL reconstructions and were enrolled in the study. (At that time, the surgeon performed 75 ligament reconstructions, which means that 85% of patients were included in the study.) Patients underwent a primary arthroscopically assisted ACL reconstruction with graft randomization into a patellar or hamstring tendon group according to operative registration list position (even number = patellar tendon, odd number = hamstring tendon). In 32 patients, ACL reconstruction was performed with doubled semitendinosus and gracilis tendon autograft (STG group) while in the other 32 patients, ACL reconstruction was performed with patellar tendon autograft (PT group). All patients signed an informed consent form. The study was approved by the National Ethical Committee.

Operative Technique

All procedures were performed by the first author (M.S.). Apart from the graft harvesting, the surgical technique was identical. All patients were examined under anesthesia (Lachman, drawer, pivot-shift, and rotation tests were documented), after which routine diagnostic arthroscopy and meniscal surgery were performed, followed by the ACL reconstruction. The femoral tunnels were drilled at the posterolateral part, while the tibial tunnels were drilled through the central part of the anatomic footprints. The femoral tunnel was drilled through the anteromedial portal before tibial tunnel drilling. In the STG group, the surgical procedure was an arthroscopic single-incision technique with double-looped semitendinosus and gracilis tendons. Drill guides were used to confirm the correct position of the tunnels. To find the femoral entry point, the surgeon used the "bull's eye" drill guide (Linvatec, Largo, Florida). Tunnel size was matched to the cross-sectional size of the graft. A marking suture using No. 0 absorbable suture was set 2.5 cm from the femoral end of the graft to ensure good entry of the graft in the tunnel and to prevent the graft from twisting around the screw during insertion. The graft was inserted via the tibial tunnel into a blind femoral tunnel. The left thread or right thread round-head, cannulated interference screw (RCI, ART-MAM, Jesenice, Slovenia) was used for femoral graft fixation.[28] In right knees, the screw with left thread was used and in left knees, the screw with right thread was used. In the right knee, the left thread screw turns in the left direction and pushes the transplant away from the lateral wall of the intercondylar notch, and vice versa. This prevented graft impingement against the lateral wall of the intercondylar notch. Tibial anatomic joint-line fixation was achieved by a bioabsorbable interference screw (Linvatec) in an outside-in direction at a knee flexion angle of approximately 10° and manual pretension of the graft.

In the patellar tendon group, the surgical procedure was an arthroscopic single-incision technique with the central third of the ipsilateral bone–patellar tendon–bone used as a free autograft with matching drill tunnels made in the femur and tibia. The defects of the patella and the proximal tibia were not bone-grafted. Tunnel size in the patellar tendon group was made 1 mm larger than the bone block size. The fixation of the graft in the drill tunnels was performed proximally with a metal RCI screw and distally with a bioabsorbable interference screw.

The average diameter of the bone tunnels in the STG group was 8 mm (range, 7-9 mm) while in the PT group it was 10 mm (range, 9-11 mm).

Rehabilitation

All the patients were rehabilitated according to the same accelerated protocol, which permitted immediate full weightbearing and full range of motion.[29] A rehabilitation brace was used for 3 weeks postoperatively. The splint was not worn during rehabilitation protocol or during the night. The importance of reaching full extension was emphasized from the beginning. The patients had to obtain an arc of at least 0° to 90° before they were discharged from the hospital. Closed kinetic chain exercises were started immediately after the operation. The leg extension exercise against resistance in the arc from 45° to 0° was not allowed during the initial 6 postoperative weeks. Running was allowed at 8 weeks and contact sports at 6 months under 4 conditions: (a) no effusion, (b) full range of motion, (c) stable knee, and (d) obtained muscle strength of 90% compared with the contralateral side.

Follow-up Evaluation

Clinical evaluations of knee function and stability and radiographic evaluation were assessed preoperatively, at 2 weeks, 6 weeks, 3 and 5 months after surgery, and at 5-year follow-up. At 11-year follow-up, the same evaluations were performed along with completion of the Short Form–36 version 2 questionnaires (Standard Form).

The Lysholm knee score questionnaire[31] and Short Form–36[23] were sent to the patients by mail and completed in their home environment. The International Knee Documentation Committee (IKDC) activity level was used to assess the functional outcome.[13] Patients were evaluated using the IKDC score according to the 2000 IKDC Knee Examination Form. Clinical ligament testing was performed by means of the Lachman test, anterior drawer test, and pivot-shift test, with side-to-side differences recorded. Objective anteroposterior knee stability was determined by using the KT-1000 arthrometer (MEDmetric, San Diego, California) at manual maximum tension and a knee flexion angle of 25°.[9] We selected 3 mm as the upper limit of normality of side-to-side difference in anterior tibial displacement. After the evaluation of the IKDC score, we graded these parameters as A (normal), B (nearly normal), C (abnormal), and D (severely abnormal) compared with the patients' preoperative conditions or the control knees.

The patients were classified as having subjective anterior knee pain if they registered pain with any of the following activities: (a) during or after activity, (b) during stair-walking, or (c) during squatting or kneeling. Before surgery and at 5 years and 11 years after surgery, 30° PA flexion weightbearing and weightbearing lateral radiographs were obtained. Patients with abnormal radiographs before surgery were not included in the study. Radiographic evaluation was done according to IKDC recommendations (A: normal; B: minimal changes and barely detectable joint-space narrowing; C: minimal changes and joint-space narrowing up to 50%; D: more than 50% joint-space narrowing). The single-legged hop test was used to evaluate functional performance.

Statistical Methods

In April 1999, a hospital database with all preoperative and postoperative variables was established. Statistical analysis was performed by an independent expert, not involved in the study protocol. Median (range) values are presented, except for the absolute anterior KT-1000 arthrometer laxity measurements, for which mean (range) values are presented. The unpaired t test was used for the normally distributed numerical data (KT-1000 arthrometer). A nonparametric analog was used where appropriate (Mann-Whitney test was used to compare the differences between the groups, eg, Lysholm score; Wilcoxon signed-rank test was used to compare preoperative and postoperative values, eg, preoperative Lysholm score vs postoperative Lysholm score). The χ^2 test was used to compare categorical variables (overall IKDC scores, the severity of osteoarthritis). A P value $<.05$ was considered statistically significant.

RESULTS

Of the 64 patients in the study, 52 (82%) were analyzed at the 11-year follow-up (Figure 1). Of 12 patients not analyzed, 3 were not available. One female patient from the STG group had a high-risk pregnancy at the time of follow-up and 1 patient from the PT group was not available. One female patient from the PT group went to another hospital for revision ACL surgery because of the reinjury of the knee. The remaining 9 patients were excluded because of a contralateral ACL rupture or revision ACL surgery of the knee. During the study period, 3 patients in the PT group and 2 patients in the STG group had a treated contralateral ACL rupture. Furthermore, 2 patients in the STG group and 2 patients in the PT group had reinjury of the knee, so revision ACL reconstruction in all had been performed at our hospital. A total of 52 patients, 27 patients in the STG group and 25 patients in the PT group, were clinically examined and roentgenographically evaluated at 11-year follow-up.

The comparison between the STG and the PT groups in relation to preoperative and intraoperative parameters showed that the 2 groups were comparable. None of the patients' radiographs showed evidence of knee joint osteoarthritis. There were no significant differences between the 2 groups regarding age, sex, preinjury activity level, preoperative Lysholm knee scores, or interval from injury to surgery (Table 1). The ACL was reconstructed at less than 12 weeks after injury in 4 of 27 patients (15%) in the STG group and in 4 of 25 patients (16%) in the PT group.

Meniscectomies were performed in 14 of 27 STG patients (52%) and in 16 of 25 PT patients (64%) (P = .413). There were significant differences between the 2 groups according to subtotal meniscectomies (P = .027), with greater

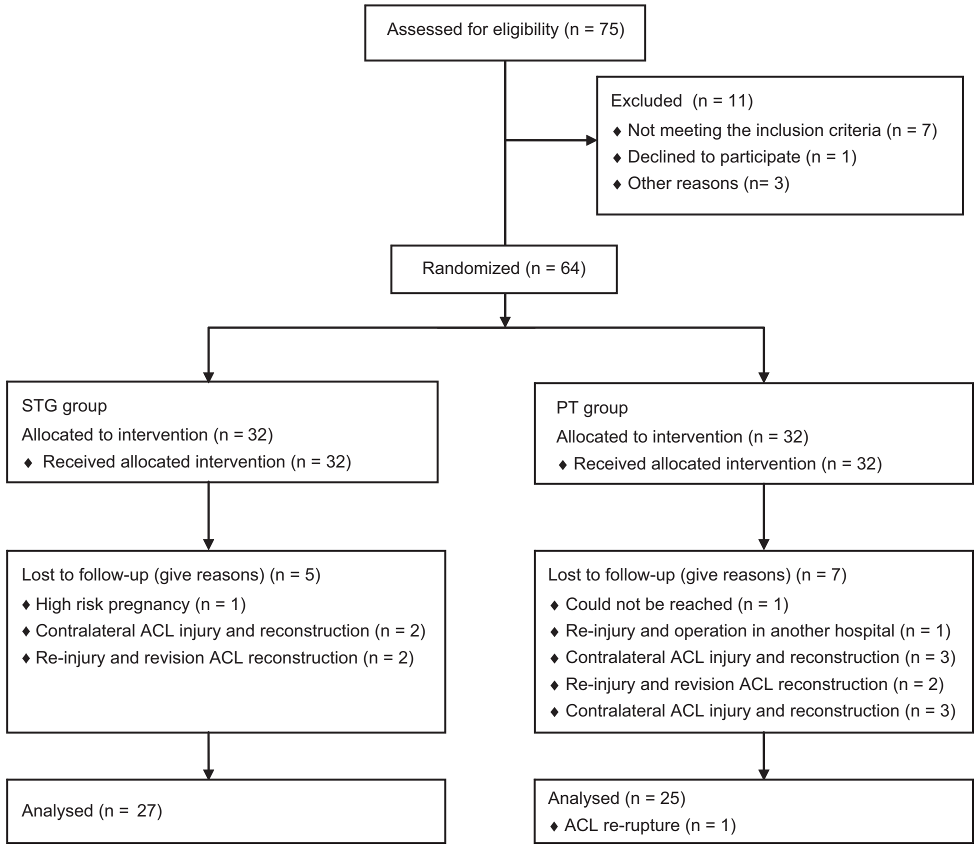


Figure 1. Patient sampling and selection.

TABLE 1
Comparison of Randomized Groups

Variable	Hamstring Tendon Group	Patellar Tendon Group	*P* Value
Total No. of patients	27	25	
Mean age at follow-up (range), y	36 (25-54)	38 (27-58)	
Gender, No.			.413
Male	14	16	
Female	13	9	
International Knee Documentation Committee (IKDC) preinjury activity level			
Level 1 (jumping and pivoting)	19	16	.878
Mean preoperative Lysholm score (range)	57 (26-82)	50 (19-86)	.187
Mean time from injury to surgery (range), mo	25 (1-84)	23 (1-60)	
Acute reconstruction	4 (14%)	4 (15%)	
Chronic reconstruction	23 (86%)	21 (85%)	
Meniscal surgery	16 (59%)	17 (68%)	.484
Meniscal repair	2 (12%)	1 (6%)	
Partial meniscectomy	6 (38%)	12 (71%)	
Subtotal meniscectomy	8 (50%)	4 (23%)	.027

prevalence in the hamstring tendon (STG) group. The number of meniscectomies is high because of the long interval from injury to surgery and very high sports activity level of the patients. The menisci were sutured in 2 patients from the STG group (12%) and in 1 patient from the PT group (6%). The arthroscopic meniscal repair procedures were performed with nonabsorbable sutures with an inside-out technique. There were no infections, deep venous thromboses, nerve injuries, or other operative complications in this series. The median length of hospital stay was 3 nights (range, 2-4) for the STG group and 4 nights (range, 2-6) for the PT group. All patients were comfortable with the postoperative rehabilitation protocol. Full extension was achieved during the early postoperative rehabilitation

TABLE 2
Lysholm Knee Score 11 Years Postoperatively

Lysholm Score	Hamstring Tendon Group (N = 27) No. of Patients (%)	Patellar Tendon Group (N = 25) No. of Patients (%)	*P* Value
Excellent (95-100)	18 (67)	14 (56)	.314
Good (84-94)	9 (33)	11 (44)	
Fair (65-83)	0 (0)	0 (0)	
Poor (<65)	0 (0)	0 (0)	

TABLE 3
Outcomes of Clinical Evaluation 11 Years Postoperatively

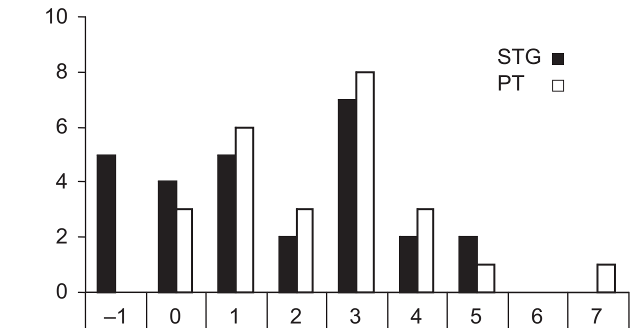
	Hamstring Tendon Group (N = 27) No. of Patients (%)	Patellar Tendon Group (N = 25) No. of Patients (%)	*P* Value
Manual Lachman test			.112
A (negative)	23 (85)	16 (64)	
B (positive with firm end point: 1+)	4 (15)	8 (32)	
C (soft end point: 2+)	0 (0)	1 (4)	
Manual pivot shift			
A (negative)	25 (93)	17 (68)	
B (glide: 1+)	2 (7)	7 (28)	.036
C (clunk: 2+)	0 (0)	1 (4)	
International Knee Documentation Committee (IKDC) instrumental anteroposterior translation (KT-1000 arthrometer, 25° flexion; manual maximum)			
A (<3 mm)	24 (89)	18 (72)	.317
B (3-5 mm)	3 (11)	6 (24)	
C (6-10 mm)	0 (0)	1 (4)	

period. There were no signs of arthrofibrosis or impingement syndrome, so further surgeries were not required. One patient from the STG group was injured during a judo national championship. An arthroscopic procedure was performed and an intact ACL was found; however, partial resection of a torn medial meniscus had to be done. Another patient from the STG group had an injury during soccer activity. During the arthroscopic procedure, we found a partial meniscal lesion and full-thickness chondral lesion, so we performed partial meniscectomy and microfracturing. At 11-year follow-up, in the STG group, 2 patients (6%) had graft ruptures and 2 patients (6%) had ruptured their contralateral ACLs. In the PT group, there were 4 cases (12%) of graft ruptures (1 did not have a revision surgery), and 3 patients (9%) had ruptured their contralateral ACLs. There was no significant difference between the STG and PT groups in the rate of ACL graft rupture (P = .673) or contralateral ACL rupture.

Preoperative versus postoperative Lysholm knee scores showed a significant improvement in both groups (STG: $P < .001$; PT: $P < .001$). At the 11-year follow-up, there were no statistically significant differences between the groups with respect to Lysholm knee scores (Table 2). In the STG group, the mean Lysholm score was 95 (range, 84-100) and in the PT group, the mean score was 94 (range, 84-100). Twenty-two patients (81%) in the STG group and 18 patients (72%) in the PT group were still at their preinjury level of activity (P = .285). The single-legged hop test of knee function determines the ratio of distance achieved by hopping on the involved limb compared with the contralateral normal limb. A grade A hop on the involved side is a distance equal to or greater than 90% of that achieved with the contralateral limb. At 11 years, 21 patients (78%) of the STG group and 17 patients (68%) of the PT group had achieved a grade A hop (P = .536).

Patients reported anterior knee pain during strenuous activities or in kneeling-squatting position in 30% of the STG group (8 of 27) and in 48% of the PT group (12 of 25) (P = .255). The range of motion of the reconstructed knee was physiologically comparable with the healthy contralateral limb in both groups, except in 1 patient (4%) from the PT group in whom a 15° lack of terminal passive flexion was measured. We did not find any loss of extension in either group.

On clinical examination at 11-year follow-up, there was a positive pivot-shift test (2+) in 1 patient in the PT group (Table 3). The graft rupture occurred 3 months before final follow-up during sports activity. Positive pivot-shift test (1+) was significantly more frequent in the PT group (P = .036). Postoperative anterior laxity values measured with the KT-1000 arthrometer, grouped as proposed by the IKDC (A, 0-2 mm; B, 3-5 mm; C, 6-10 mm; D, >10 mm) did not result in a significant difference. The mean value of anterior laxity measured with the KT-1000 arthrometer at manual maximum tension (side-to-side difference) was 1.5 ± 2.0 mm for the STG group



	–1	0	1	2	3	4	5	6	7
■ STG	5	4	5	2	7	2	2	0	0
□ PT	0	3	6	3	8	3	1	0	1

Figure 2. Comparison of anterior laxity at 11 years after surgery measured with the KT-1000 arthrometer (side-to-side difference in millimeters, at manual maximum tension). The mean value was 1.5 ± 2.0 mm for the hamstring tendon (STG) group and 2.5 ± 1.7 mm for the patellar tendon (PT) group (*P* = .069).

and 2.5 ± 1.7 mm for the PT group (*P* = .069). In the STG group, 89% (24 of 27) scored 3 mm or less difference, as did 72% (18 of 25) of the patellar tendon group (*P* = .317). Two percent of patients (1 of 52) scored more than 5 mm. Figure 2 represents all the KT-1000 side-to-side difference measurements performed at 11-year follow-up. One patient in the PT group had abnormal KT results (7 mm) that correlated with the clinical examination and the patient's subjective opinions. Exclusion of negative or abnormal KT-1000 arthrometer results from analysis does not significantly affect the mean value of KT-1000 arthrometer measurements. Comparison of objective anteroposterior knee stability associated with the patients' gender indicated significant differences. We found that the mean manual maximum side-to-side difference for men was 1.5 ± 1.8 mm and for women, 2.6 ± 1.9 mm (*P* = .030).

As stated previously, the IKDC system was used for grading radiographs. The medial, lateral, and patellofemoral compartments were examined for evidence of joint-space narrowing and for the presence of osteophytes. The worst compartment grading was used as the overall grade. Table 4 represents that radiographic evidence of knee joint osteoarthritis grade B, C, or D was present in 84% of patients (21 of 25) in the PT group and in 63% of patients (17 of 27) in the STG group, which was statistically significant (*P* = .008). No significant correlation was found between the grade of osteoarthritis and meniscectomy in either group (STG group, *P* = .14; PT group, *P* = .54). There was no correlation between the grade of osteoarthritis and the KT-1000 arthrometer side-to-side difference measurements (STG group, *P* = .09; PT group, *P* = .40). However, a moderate correlation was found between the grade of osteoarthritis and the follow-up level of activity in the STG group (Spearman rho = .61; $P < .01$). No correlation existed between osteoarthritis and the level of activity at the follow-up in the PT group (Spearman rho = .14; *P* = .52). One patient from the STG group with grade D generalized

TABLE 4
IKDC Grading of Degenerative Joint Disease Comparing STG and PT Groups[a] (*P* = .008)

Knee Joint Arthritis Grade	STG Group,[b] No. of Patients	PT Group,[c] No. of Patients
A	10	4
B	14	10
C	2	11
D	1	0

[a]IKDC, International Knee Documentation Committee; STG, hamstring tendon; PT, patellar tendon.
[b]17 of 27 patients (63%).
[c]21 of 25 patients (84%).

osteoarthritis has been a professional judo athlete with many injuries of the reconstructed knee. At final follow-up, he had full knee extension but a 15° flexion deficit; the knee was objectively stable, but painful during sports activities.

The IKDC score was based on the original IKDC evaluation method,[13] that is, it was not changed during the study period, considering the recent development of a questionnaire relating to "subjective" factors, to ensure uniformity of reporting during the 11-year period. Table 5 represents the distribution of patients according to overall IKDC score. Overall IKDC evaluation grade A was obtained in 59% (16 of 27) of patients in the STG group and in 32% (8 of 25) of patients in the PT group. There was no statistically significant difference between groups (*P* = .071).

There were no differences between groups in health perception and health-related quality of life. Groups were comparable in all parameters of the Short Form–36 version 2 questionnaire (Figure 3). Normalized Physical Component Summary (PCS) was 55.9 in the PT group, and 52.5 in the STG group (*P* = .134). Normalized Mental Component Summary (MCS) was 50.3 in the PT group, and 52.1 in the STG group (*P* = .234).

DISCUSSION

In a randomized controlled trial, we compared the use of the well-established and the most frequently used central-third bone–patellar tendon–bone autograft with use of a doubled semitendinosus and gracilis tendon autograft. Apart from the graft, all other important factors for the clinical outcome, such as fixation technique and the rehabilitation protocol, were identical. One surgeon performed all of the reconstructions, and unbiased observers, not involved in the operation or rehabilitation, performed the postoperative evaluations. The strength of this study is the fact that this is one of the longest known follow-up periods among controlled prospective comparative studies. Numerous studies have recently been published comparing patellar and hamstring tendon autografts in arthroscopic ACL reconstruction.[5,15,18] The present study continues to provide ongoing data showing the similarities and differences in the clinical results between the 2 groups.

TABLE 5
Overall IKDC Score[a] (P = .071)

IKDC Score	STG Group No. of Patients (%)	PT Group No. of Patients (%)
A (normal)	16 (59)	8 (32)
B (nearly normal)	11 (41)	16 (64)
C (abnormal)	0 (0)	1 (4)
D (severely abnormal)	0 (0)	0 (0)

[a]IKDC, International Knee Documentation Committee; STG, hamstring tendon; PT, patellar tendon.

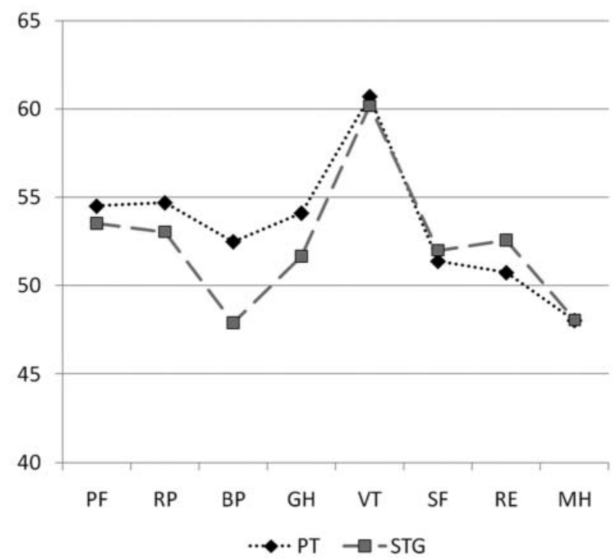


Figure 3. Mean Short Form–36 version 2 normalized subscores at the 11-year follow-up in the patellar tendon group compared with the hamstring tendon (STG) group. Corresponding P values: PF (Physical Functioning, P = .75), RP (Role Physical, P = .62), BP (Bodily Pain, P = .19), GH (General Health, P = .44), VT (Vitality, P = .75), SF (Social Functioning, P = .88), RE (Role Emotional, P = .14), MH (Mental Health, P = .91).

The results of our study substantiate the similarities found in previous reports that document that good or excellent results may be obtained in the majority of ACL reconstructions when using either PT or STG autografts. A meta-analysis of Biau at al[6] concluded that ACL reconstruction with PT autograft was significantly associated with a decreased risk of a positive pivot-shift test result (P = .016). However, our study shows the opposite, that there is a statistically significant difference according to positive pivot-shift test (1+) in the PT group (P = .044). On the other hand, our study shows that there is a statistically significant difference according to the radiographic changes, possibly pointing to the grade B and C osteoarthritis in patients from the PT group. In this study, 84% of patients in the PT group (21 of 25) and 63% of patients in the STG group (17 of 27) had radiographic evidence of grade B, C, or D osteoarthritis affecting the knee joint at 11 years after ACL reconstruction (P = .008). No significant correlation was found between the grade of osteoarthritis and meniscectomy in either group (STG group, P = .14; PT group, P = .54). Meniscectomies were performed in 16 of 27 STG patients (59%) and in 16 of 25 PT patients (64%) (P = .523), out of which 2 patients (STG group) underwent subsequent meniscectomy. There were significant differences between the 2 groups according to subtotal meniscectomy (P = .027), with higher prevalence in the STG group. This emphasizes the hypothesis that the choice of the graft is crucial in development of degenerative knee joint disease at 11 years after ACL reconstruction. Barenius et al[4] found significant influence of meniscal injury on the incidence of osteoarthritis, but without correlation with graft choice. Holm et al[14] pointed out similar results. The prevalence of osteoarthritis was significantly higher in the operated leg than in the contralateral leg, but there were no significant differences between the 2 groups. Keays et al,[16] in their 6-year follow-up study, found that reconstruction using the hamstring tendon resulted in a significantly lower incidence of osteoarthritis.

There is only 1 prospective long-term clinical study comparable with our study. Pinczewski et al[26] evaluated 2 groups of 90 patients each, consecutively treated with hamstring or patellar tendon grafts at 10 years after ACL reconstruction. In contrast to our study, in which all meniscectomies are included, their study eliminated all patients with more than one-third meniscectomy. In both studies, patients with contralateral ACL rupture and patients who required revision ACL surgery of the knee during the follow-up period were excluded. Both studies state the overall rate of ACL ruptures and neither of the studies found a significant difference in comparing the 2 different autografts. In distinction to our methods, the tibial graft fixation in their study was performed with metal RCI screws. Because of the fact that the screw is made of metal and not absorbable material, it is not appropriate to achieve the fixation of the graft at the tibial anatomic joint line. The removal of a metal screw, when revision ACL surgery is required, could be a technically demanding procedure. Pinczewski et al found that patients with patellar tendon grafts had a greater prevalence of osteoarthritis at 10 years after surgery (P = .040), which supports results of our study.

Extension deficit was determined as the loss of extension in the involved limb in comparison with the passively maximally extended posture of the contralateral uninjured or "normal" limb. Historically, limitation of extension had been a significant problem after ACL reconstruction when using the central third of the patellar tendon. Early postoperative mobilization with emphasis on full passive extension has minimized extension loss. Pinczewski et al reported that 31% of the patellar tendon group had a fixed flexion deformity and 19% of the hamstring tendon group had fixed flexion deformity at 5-year follow-up. Our results demonstrate no statistically significant difference in range of motion after hamstring tendon reconstruction compared with patellar tendon reconstruction. This study also confirms that harvesting the semitendinosus and gracilis tendons or the middle third of the patellar tendon does not cause a permanent extension loss.

Comparing the anterior knee pain between the groups, we did not find a significant difference, as many other studies have not.[4,14,18] In previous studies, donor-site problems have been described after ACL reconstruction with bone–patellar tendon–bone autografts. Loss of full range of motion and disturbances in knee sensitivity have been proposed as factors correlating with subjective anterior knee pain and discomfort.[20] In our opinion, donor-site symptoms do occur, but clear documentation of their origin is difficult because of problems with differentiating between the types of symptoms experienced. As determined by the IKDC evaluation, the number of donor-site symptoms in any form was minimal in both groups and not statistically significant. Kneeling pain was reported if it was present after patients kneeled on the carpeted floor. Patients reported anterior knee pain during strenuous activities, or in kneeling-squatting position, in 30% in the STG group (8 of 27) and in 48% in the PT group (12 of 25) (P = .255). Pinczewski et al[25] reported significant differences in kneeling pain comparing hamstring and patellar tendon groups, where the kneeling pain was greater in the patellar tendon group. In our study, these differences were not significant.

One of the key points of this study was the evaluation of objective stability after the reconstruction. At 11 years' follow-up, objective anteroposterior knee stability was measured by using the KT-1000 arthrometer. We performed the KT-1000 arthrometer measurements with manual maximum tension. A force of 134 N is often insufficient, particularly in muscular or heavy patients. In these patients, the manual maximum test gives more stringent results.[1] Most investigations showed better static stability when a patellar tendon graft was used.[10,11,25] However, Wagner et al[32] found significantly fewer positive pivot-shift test results in the hamstring tendon group (P = .005). The KT-1000 arthrometer side-to-side difference was 2.6 mm for the patellar tendon group and 2.1 mm for the hamstring tendon group (P = .041). In our study, there was no significant difference between groups with respect to the Lachman test, but our study shows that there is a statistically significant difference according to positive pivot-shift test (1+) in the PT group at 11 years of follow-up (P = .044). The mean value of objective anterior laxity measurements revealed no significant differences between the STG and the PT groups (P = .069). Ferrari et al[12] compared clinical results between men and women after ACL reconstruction with the patellar tendon autograft. They did not find significant differences with respect to the Lachman and pivot-shift tests or failure rate. However, they did find that the mean maximum side-to-side difference for men was 0.76 ± 2.8 mm and for women was 1.73 ± 2.2 mm, which is a statistically significant difference (P = .014). In our study, we found that the mean manual maximum side-to-side difference for men was 1.5 ± 1.8 mm and for women, 2.6 ± 1.9 mm (P = .030).

The overall results of the present study, as measured by the IKDC evaluating system, were 100% normal or nearly normal in the STG group and 96% in the PT group. This result is high in comparison with the results of other prospective studies.[4,14,16,18,26] The overall IKDC results in the present study may be so high because of the limited number of patients. The other limitation in the design of our study is that there were meniscal injuries as a combined lesion to the ACL tear. It is difficult to obtain patients with solitary ACL tears without any other intra-articular lesions. Unfortunately, the randomization is not considered an ideal method as it is not completely hidden. Two unbiased observers performed the clinical examination at 11-year follow-up, but they were not blinded for the skin scars.

We could not find any differences between groups in terms of health-related quality of life, as measured by the Short Form–36 version 2 questionnaire. There is 1 study that compared the quality of life at long-term follow-up after ACL reconstruction using the patellar tendon graft or quadrupled semitendinosus graft. Barenius et al[4] also did not find any differences between groups, or significant differences to an age-matched healthy Swedish population.

The present study is a follow-up of a 5-year follow-up study.[28] Comparing the results, we found that a positive pivot-shift test (1+) was significantly more frequent in the PT group at 11-year follow-up. Both studies found that patients with PT grafts had a greater prevalence of osteoarthritis.

In conclusion, the results of our study substantiate the similarities found in previous reports that document that good or excellent results may be obtained in the majority of ACL reconstructions when using either PT or STG autografts. Both hamstring and patellar tendon grafts provided good subjective outcomes and objective stability at 11 years. Positive pivot-shift test (1+) has been significantly more frequent in the PT group. No significant differences in the rate of graft failure or contralateral ACL rupture were identified. Patients with PT grafts had a greater prevalence of osteoarthritis at 11 years after surgery.

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REFERENCES

1. Aglietti P, Buzzi R, Zaccherotti G, De Biase P. Patellar tendon versus doubled semitendinosus and gracilis tendons for anterior cruciate ligament reconstruction. *Am J Sports Med*. 1994;22:211-218.
2. Bach BR Jr, Jones GT, Sweet FA, Hager CA. Arthroscopy-assisted anterior cruciate ligament reconstruction using patellar tendon substitution: two- to four-year follow-up results. *Am J Sports Med*. 1994;22:758-767.
3. Bach BR Jr, Tradonsky S, Bojchuk J, Levy ME, Bush-Joseph CA, Khan NH. Arthroscopically assisted anterior cruciate ligament reconstruction using patellar tendon autograft: five- to nine-year follow-up evaluation. *Am J Sports Med*. 1998;26:20-29.
4. Barenius B, Nordlander M, Ponzer S, Tidermark J, Eriksson K. Quality of life and clinical outcome after anterior cruciate ligament reconstruction using patellar tendon graft or quadrupled semitendinosus

graft: an 8-year follow-up of a randomized controlled trial. *Am J Sports Med*. 2010;38:1533-1541.
5. Beynnon BD, Johnson RJ, Fleming BC, et al. Anterior cruciate ligament replacement: comparison of bone-patellar tendon-bone grafts with two-strand hamstring grafts: a prospective, randomized study. *J Bone Joint Surg Am*. 2002;84:1503-1513.
6. Biau JB, Katsahian S, Kartus J, et al. Patellar tendon versus hamstring tendon autografts for reconstructing the anterior cruciate ligament : a meta-analysis based on individual patient data. *Am J Sports Med*. 2009;37(12):2470-2478.
7. Bonamo JJ, Krinick RM, Sporn AA. Rupture of the patellar ligament after use of its central third for anterior cruciate reconstruction: a report of two cases. *J Bone Joint Surg Am*. 1984;66:1294-1297.
8. Coombs R, Cochrane T. Knee flexor strength following anterior cruciate ligament reconstruction with the semitendinosus and gracilis tendons. *Int J Sports Med*. 2001;22:618-622.
9. Daniel DM, Malcolm LL, Losse G, Stone ML, Sachs R, Burks R. Instrumented measurement of anterior laxity of the knee. *J Bone Joint Surg Am*. 1985;67:720-726.
10. Ejerhed L, Kartus J, Sernert N, Kohler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med*. 2003;31:19-25.
11. Eriksson K, Anderberg P, Hamberg P, et al. A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br*. 2001;83:348-354.
12. Ferrari JD, Bach BR Jr, Bush-Joseph CA, Wang T, Bojchuk J. Anterior cruciate ligament reconstruction in men and women: an outcome analysis comparing gender. *Arthroscopy*. 2001;17:588-596.
13. Hefti F, Muller W, Jakob RP, Staubli HU. Evaluation of the knee ligament injuries with the IKDC form. *Knee Surg Sports Traumatol Arthrosc*. 1993;1:226-234.
14. Holm I, Øiestad BE, Risberg MA, Aune AK. No difference in knee function or prevalence of osteoarthritis after reconstruction of the anterior cruciate ligament with 4-strand hamstring autograft versus patellar tendon–bone autograft: a randomized study with 10-year follow-up. *Am J Sports Med*. 2010;38:448-454.
15. Jansson KA, Linko E, Sandelin J, Harilainen A. A prospective randomized study of patellar versus hamstring tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2003;31:12-18.
16. Keays SL, Bullock-Saxton JE, Keays AC, Newcombe PA, Bullock MI. A 6-year follow-up of the effect of graft site on strength, stability, range of motion, function, and joint degeneration after anterior cruciate ligament reconstruction: patellar tendon versus semitendinosus and gracilis tendon graft. *Am J Sports Med*. 2007;35:729-739.
17. Kurosaka M, Yoshiya S, Andish JT. A biomechanical comparison of different surgical techniques of graft fixation in anterior cruciate ligament reconstruction. *Am J Sports Med*. 1987;15:225-229.
18. Liden M, Ejerhed L, Sernert N, Laxdal G, Kartus J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction: a prospective, randomized study with a 7-year follow-up. *Am J Sports Med*. 2007;35:740-748.
19. Maeda A, Shino K, Horibe S, Nakata K, Buccafusca G. Anterior cruciate ligament reconstruction with multistranded autogenous semitendinosus tendon. *Am J Sports Med*. 1996;24(4):504-509.
20. Marder RA, Raskind JR, Carroll M. Prospective evaluation of arthroscopically assisted anterior cruciate ligament reconstruction: patellar tendon versus semitendinosus and gracilis tendon. *Am J Sports Med*. 1991;19:478-484.
21. Marumoto JM, Mitsunaga MM, Richardson AB, Medoff RJ, Mayfield GW. Late patellar tendon ruptures after removal of the central third for anterior cruciate ligament reconstruction: a report of two cases. *Am J Sports Med*. 1996;24:698-701.
22. Mastrokalos DS, Springer J, Siebold R, Paessler HH. Donor site morbidity and return to the preinjury activity level after anterior cruciate ligament reconstruction using ipsilateral and contralateral patellar tendon autograft. *Am J Sports Med*. 2005;33:85-93.
23. McAllister DR, Motamedi AR, Hame SL, Shapiro MS, Dorey FJ. Quality of life assessment in elite collegiate athletes. *Am J Sports Med*. 2001;29:806-810.
24. Northrup T, Lintner D, Farmer J, et al. Biomechanical evaluation of interference screw fixation of hamstring and patellar tendon grafts used in ACL reconstruction. *Orthop Trans*. 1997;21:174.
25. Pinczewski LA, Deehan DJ, Salmon LJ, Russell VJ, Clingeleffer A. A five-year comparison of patellar tendon versus four-strand hamstring tendon autograft for arthroscopic reconstruction of the anterior cruciate ligament. *Am J Sports Med*. 2002;30:523-536.
26. Pinczewski LA, Lyman J, Salmon LJ, Russell VJ, Roe J, Linklater J. A 10-year comparison of anterior cruciate ligament reconstructions with hamstring tendon and patellar tendon autograft. *Am J Sports Med*. 2007;35(4):564-574.
27. Rowden NJ, Sher D, Rogers GJ, Schindhelm K. Anterior cruciate ligament graft fixation: initial comparison of patellar tendon and semitendinosus autografts in young fresh cadavers. *Am J Sports Med*. 1997;25:472-478.
28. Sajovic M, Vengust V, Komadina R, Tavcar R, Skaza K. A prospective, randomized comparison of semitendinosus and gracilis tendon versus patellar tendon autografts for anterior cruciate ligament reconstruction: five-year follow-up. *Am J Sports Med*. 2006;34:1933-1940.
29. Shelbourne KD, Nitz P. Accelerated rehabilitation after anterior cruciate ligament reconstruction. *Am J Sports Med*. 1990;18:292-299.
30. Steiner ME, Hecker AT, Brown CH Jr, Hayes WC. Anterior cruciate ligament graft fixation: comparison of hamstring and patellar tendon grafts. *Am J Sports Med*. 1994;22:240-247.
31. Tegner Y, Lysholm J. Rating systems in the evaluation of the knee ligament injuries. *Clin Orthop Relat Res*. 1985;198:43-49.
32. Wagner M, Kääb MJ, Schallock J, Haas NP, Weiler A. hamstring tendon versus patellar tendon anterior cruciate ligament reconstruction using biodegradable interference fit fixation: a prospective matched-group analysis. *Am J Sports Med*. 2005;33:1327-1336.
33. West RV, Harner CD. Graft selection in anterior cruciate ligament reconstruction. *J Am Acad Orthop Surg*. 2005;13:197-207.
34. Williams RJ III, Hyman J, Petrigliano F, Rozental T, Wickiewicz TL. Anterior cruciate ligament reconstruction with a four-strand hamstring tendon autograft. *J Bone Joint Surg Am*. 2004;86:225-232.